ASSIGNMENT

Basic Concepts of Full Stack Web Development

Beginner Level • Theory and Practical Planning

Student Name	_____
Student ID	_____
Submission Date	_____

Assignment Overview

In this assignment, you will learn how the front end, back end, and database work together to create a web application. You will explain key terms, identify common technologies, and plan a simple website suitable for a beginner.

Learning Outcomes

By completing this assignment, you should be able to:

- Explain what full stack web development means.
- Distinguish between front-end and back-end development.
- Describe the purpose of HTML, CSS, JavaScript, a server-side language, and a database.
- Explain how a browser, server, and database communicate.
- Plan a small web application and describe its basic features.

General Instructions

- Answer all questions in your own words.
- Use examples where possible. Simple diagrams are encouraged.
- No advanced programming is required for this assignment.
- If you use online sources, list their titles and links at the end.
- Submit your work as a PDF or a neatly formatted document, unless your instructor specifies another format.

Part A — Understanding the Basics

Answer each question using approximately 2–5 sentences unless a different format is requested.

1. What is a website? Give one example of a website you use.

2. What is full stack web development? Explain what the word “stack” refers to.

3. Explain the difference between front-end development and back-end development.

4. What is the purpose of each technology below?
a) HTML b) CSS c) JavaScript d) A server-side language (for example, PHP, Java, or Python) e) A database

5. Explain the difference between a web browser, a web server, and a database.

6. What is an API? Describe one reason a front end might need to communicate with a back end through an API.

7. What is the difference between a static website and a dynamic website? Give one example of each.

Part B — How a Full Stack Application Works

Imagine a student registration website. A student enters their name and email address into a form and presses “Register.”

8. Describe the steps that happen from the moment the student presses the Register button until the website displays a result. Include the browser/front end, request, back end, database, and response.

9. Draw a simple labelled diagram showing how the following components communicate:
User/Browser → **Front End** → **Back End/API** → **Database**
Add arrows to show the request and response where appropriate.

10. Why should a web application validate user input? Give two examples of checks that could be performed on a registration form.

Part C — Mini Project Planning

Plan a simple beginner-friendly web application. Choose one idea: a personal portfolio, a to-do list, a student registration page, or a small library book list.

11. Give your project a name and explain its purpose in 3–4 sentences.

12. List at least three features the user can access. For each feature, say what the user can do.

13. Identify which parts belong to the front end, which tasks would be handled by the back end, and what information (if any) should be saved in a database.

14. Choose suitable beginner technologies for your project and explain your choices. You may select HTML, CSS, JavaScript, PHP, Java/Spring Boot, Python, and MySQL or another database.

15. Sketch the main page of your application on paper or with a simple digital tool. Label the page title, navigation (if needed), main content, and at least one interactive element.

Submission Checklist

Before submitting, check that you have:

- ■ Answered questions 1–15.
- ■ Included a labelled diagram for Question 9.
- ■ Included a simple page sketch for Question 15.
- ■ Explained your technology choices.
- ■ Added source links if you used external materials.
- ■ Included your name and submission date.

Marking Guide (100 Marks)

Assessment Area	Marks
Basic concepts and terminology (Questions 1–7)	30
Application workflow and diagram (Questions 8–10)	25
Mini project plan and technology choices (Questions 11–15)	30
Clarity, organisation, and completeness	10
Sources and presentation	5
Total	**100**

Suggested Assignment Description for Your LMS

Title: Basic Concepts of Full Stack Web Development

Description: This beginner-level assignment introduces the core concepts of full stack web development. Students will explore the roles of front-end and back-end development, learn the purpose of common web technologies and databases, describe how data moves through a web application, and plan a small web project. No advanced coding experience is required. Students should answer all questions in their own words and include a simple system diagram and a page sketch.

Suggested instructions to display in the LMS: Complete all sections and submit your answers as a PDF or document. Include your name, a labelled diagram, and your mini-project plan. Use clear explanations and examples. Cite any external sources you use.

Instructor note: You can add your own due date, submission rules, and course-specific requirements before publishing this assignment.